

Controlling Cup-Ball: How the Body Helps the Brain

Meeting Notes — March 2026, Dipankar Bhattacharya

The Shared Experimental Platform

Task: transport a cup of coffee (Cup-ball (cart-pendulum) proxy) along a horizontal track using a haptic manipulandum.

Challenge: the pendulum is *unactuated*—the brain controls only the cup position, not the ball inside.

Probe: impulsive perturbations (assistive / resistive / unexpected) are injected mid-motion to reveal the control strategy.

Why it matters: mirrors everyday manipulation of underactuated objects (cups, tools, bags).

Three papers, one platform

- **Bazzi 2018** — stability analysis
- **Razavian 2021** — zero-force trajectory
- **Razavian 2023** — body mechanics + OFC

Each paper probes a *different* facet of the same question:
How do humans tame complex internal dynamics?

Core idea: identify regions of state space where the cart–pendulum dynamics are *self-converging* (contraction regions), then test whether humans steer through them.

Why not Lyapunov? Lyapunov requires proximity to a fixed equilibrium. Transport tasks begin and end *far* from equilibrium $\Rightarrow$ contraction analysis is the right tool.

Key findings:

- **Assistive** perturbation — trajectories enter a contraction region *before* the perturbation; convergent dynamics absorb the disturbance.
- **Resistive** perturbation — trajectories enter a contraction region only *after* the perturbation (reactive re-stabilisation).

Take-home

Humans exploit the *task's own dynamics* to stabilise movement. By traversing contraction regions they avoid chaotic amplification of errors and make their behaviour *more predictable* — without needing to minimise force or compute an optimal trajectory.

What it does NOT provide

- No explicit controller model
- No cost function

Problem: standard models (min-jerk, OFC min-effort, constant impedance) all fail when real cup-and-ball dynamics are present.

Proposed solution — three elements combined:

1. Spring–damper **impedance** between OFC reference and cup.
2. **OFC** in the feedback loop (not a pre-planned trajectory).
3. **Minimum-jerk** objective.

Zero-force trajectory: The smooth, jerk-minimising OFC reference $y_{\text{ref}}(t)$ keeps the impedance from pushing against the pendulum — it is *not* obtained by minimising F_{inter} directly.

Take-home

Impedance + OFC + minimum jerk is the *minimal* combination that reproduces human cup-and-ball behaviour.

No single element alone is sufficient.

Signal chain:

$$u \xrightarrow{\tau_m} F \xrightarrow{M_{\text{ref}}} y_{\text{ref}} \xrightarrow{k,d} F_{\text{imp}} \rightarrow \text{plant}$$

Question: if we add mechanical impedance to the body model, what can we still infer about the neural controller?

Four OFC variants: min-effort vs. min-jerk $\times$ with vs. without impedance.

Three key findings:

1. **Impedance is essential.** Reproduces force jump & velocity rebound < 20 ms — faster than any reflex.
2. **Cost function irrelevant.** With impedance, effort and jerk models fit equally well ($p \geq 0.195$).
3. **Feedback $\equiv$ feedforward.** Impedance handles catch trials; online feedback adds nothing.

Take-home

Ignoring impedance *over-credits the brain*.

The arm's passive spring-damper mechanics produce rapid disturbance rejection and make different “optimal” strategies look indistinguishable in the data.

Side-by-Side Comparison

	Bazzi 2018	Razavian 2021	Razavian 2023
Core tool	Contraction / PDE metric	Impedance-mediated OFC	Stochastic OFC + impedance
Controller?	None (analysis only)	Partial (impedance attractor)	Full LQG + muscle
Cost function	N/A	Min jerk $\Rightarrow$ ZFT	Effort vs. jerk — indistinguishable
Feedback role	Bypassed by contraction	Not analysed	Redundant when impedance present
Main finding	Humans exploit passive convergence	ZFT is key reference	Body mechanics subsumes controller

Common thread

All three papers show that *passive body mechanics* — not neural computation alone — are the decisive factor in successful complex-object manipulation.

Implications for Our Project

What we need (and have):

- ✓ Impedance controller $F_{\text{imp}} = K\Delta p + D\Delta\dot{p}$
- ✓ ZFT reference mass block
- ✓ LQR + muscle dynamics (14-D state)
- ✓ Cart-pendulum PyDrake simulation

What we do NOT need (for now):

- ✗ Contraction theory — useful for analysis, not for building the controller
- ✗ Exhaustive cost-function comparison — with impedance, both objectives perform the same

Bottom line

Impedance is the critical architectural ingredient. The LQR / OFC choice and feedback structure are secondary once impedance is in the loop.

Our Implementation: System Architecture

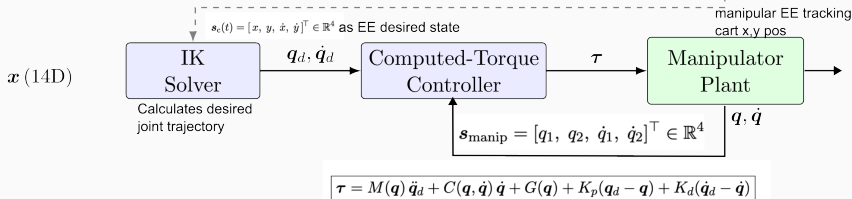
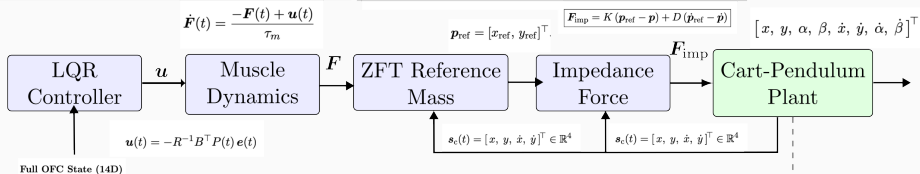
Full OFC State (14D) $\mathbf{x} = [x, y, \alpha, \beta, \dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}, F_x, F_y, x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^T \in \mathbb{R}^{14}$

$$J = \int_0^T [e^T Q e + u^T R u] dt + e(T)^T Q_N e(T)$$

$e = \mathbf{x} - \mathbf{x}^*$ is the 14D state error with goal state

$$\ddot{\mathbf{p}}_{\text{ref}}(t) = \frac{1}{M_{\text{ref}}} [\mathbf{F}(t) + K(\mathbf{p}(t) - \mathbf{p}_{\text{ref}}(t)) + D(\dot{\mathbf{p}}(t) - \dot{\mathbf{p}}_{\text{ref}}(t))]$$

$$\mathbf{p} = [x, y]^T, \mathbf{p}_{\text{ref}} = [x_{\text{ref}}, y_{\text{ref}}]^T$$



Methodology: Cart-Pendulum OFC (14-D State)

Full OFC state:

$$\mathbf{x} = [x, y, \alpha, \beta, \dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}, F_x, F_y, x_r, y_r, \dot{x}_r, \dot{y}_r]^\top \in \mathbb{R}^{14} \quad J = \int_0^T [\mathbf{e}^\top Q \mathbf{e} + \mathbf{u}^\top R \mathbf{u}] dt + \mathbf{e}(T)^\top Q_N \mathbf{e}(T)$$

Muscle dynamics ($\tau_m = 30$ ms):

$$\dot{F}_i = \frac{-F_i + u_i}{\tau_m}$$

ZFT reference mass ($M_{\text{ref}} = 1$ kg, $K = 50$, $D = 10$):

$$M_{\text{ref}} \ddot{\mathbf{p}}_{\text{ref}} = \mathbf{F} + K(\mathbf{p} - \mathbf{p}_{\text{ref}}) + D(\dot{\mathbf{p}} - \dot{\mathbf{p}}_{\text{ref}})$$

Impedance force:

$$\mathbf{F}_{\text{imp}} = K(\mathbf{p}_{\text{ref}} - \mathbf{p}) + D(\dot{\mathbf{p}}_{\text{ref}} - \dot{\mathbf{p}})$$

Finite-horizon LQR cost:

Backward Riccati:

$$-\dot{P} = A^\top P + PA - PBR^{-1}B^\top P + Q$$

Control law:

$$\mathbf{u}(t) = -R^{-1}B^\top P(t) \mathbf{e}(t)$$

Cost matrices

$Q =$

$\text{diag}(100, 100, 500, 500, 10, 10, 100, 100, 0.1, 0.1, 1, 1, 0.1, 0.1)$

$Q_N = 2Q, \quad R = \text{diag}(1, 1)$

Methodology: Manipulator Tracking

Goal: end-effector tracks cart position

$$\mathbf{p}_{\text{cart}} = [x, y]^T.$$

IK Solver (Drake InverseKinematics):

$$f_{\text{FK}}(\mathbf{q}_d) = \mathbf{p}_{\text{cart}}, \quad \|\mathbf{q}_d - \mathbf{q}_{\text{prev}}\|_2 \rightarrow \min$$

Warm-started from previous solution for continuity.

Computed-torque controller:

$$\boldsymbol{\tau} = M(\mathbf{q})\ddot{\mathbf{q}}_d + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + G(\mathbf{q}) + K_p(\mathbf{q}_d - \mathbf{q}) + K_d(\dot{\mathbf{q}}_d - \dot{\mathbf{q}})$$

Manipulator state

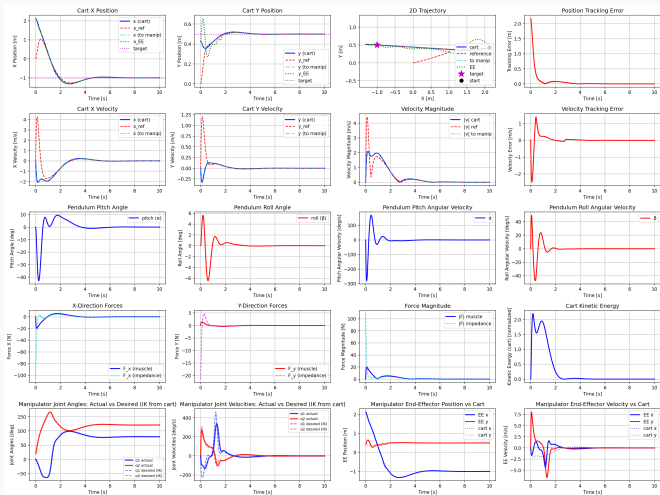
$$\mathbf{s}_{\text{manip}} = [q_1, q_2, \dot{q}_1, \dot{q}_2]^T \in \mathbb{R}^4$$

Implementation

- Framework: PyDrake MultibodyPlant
- Linearisation: automatic Jacobian (Drake API)
- Both plants share one DiagramBuilder
- Simulation step: 1 ms; horizon $T = 10\text{ s}$
- Visualisation: Meshcat 3-D (real-time)

Our Implementation: Simulation Results

Finite-horizon LQR — cup manipulator (EE tracking)



Key observations

- Cart reaches target position; pendulum angle stabilises to near-zero
- Muscle dynamics introduce a smooth ~ 90 ms force rise (3τ) — no torque spikes
- Impedance (ZFT reference mass) absorbs residual interaction forces without neural re-planning
- Manipulator end-effector follows cart via IK; joint torques remain within actuator limits